Fiscal Year:	FY 2014	**Task Last Updated:**	FY 09/24/2014
PI Name:	Cheng, Zhengdong Ph.D.		
Project Title:	Liquid Crystals of Nanoplates		
Division Name:	Physical Sciences		
Program/Discipline:	FLUID PHYSICS		
Program/Discipline--Element/Subdiscipline:	COMPLEX FLUIDS/SOFT MATTER--Complex Fluids		
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Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2013 Complex Fluids & Macromolecular Biophysics NNH13ZTT001N
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No. of PhD Candidates:	1	**No. of Master' Degrees:**	0
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No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA GRC
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Flight Program:			
Flight Assignment:	Liquid Crystal Facility		
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
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Task Description:	The central objectives of the proposed research are to investigate the dynamics of phase transitions, including the formation of liquid crystals as well as the equilibrium gel or empty liquid state, of highly anisotropic colloidal disks. We offer unprecedented control over the size (20 nm to 1 micron), thickness (a few to hundreds of layers), and their polydispersities (14% to 30%) of these disks via the controlled growth of layered inorganic crystals. The nanoplates obtained via exfoliation of the crystals have identical thickness of the order of 1 nm and controlled size and size distribution. We plan to use the ACE microscope in microgravity environment to observe the kinetics of liquid crystal (the nematics, chiral nematics, smectics, and the twist grain boundary phase) nucleation and growth, and kinetics of gelation phase separation. The comparison studies on the growth of the chiral crystals in 1g and mciro-g will reveal the role of gravity in alignment of the liquid crystal domains. The proposed research extends the existing ACE investigations		

	to understand the full phase transitions of discotic colloids by the capability to fabricate disks/nanoplates with controlled aspect ratios. It will offer novel colloidal self-assembly structures including the chiral nematics, the smectics, the twist grain boundary (TGB) phase, and the equilibrium gel or empty liquid state. The understanding of the structure and properties of colloidal disks are important to soil science and space exploration (e.g., Mars), to the fundamental investigation of discotic liquid crystals, as well as various industries, including oil recovery, the delivery of pharmaceuticals, nutrition, and cosmetics.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The understanding of the structure and properties of colloidal disks are important to soil science and space exploration (e.g., Mars), to the fundamental investigation of discotic liquid crystals, as well as various industries, including oil recovery, the delivery of pharmaceuticals, nutrition, and cosmetics.
Task Progress:	1. Phase behaviors of charged and non-charged nanoplates We planned to study the kinetics of nucleation and growth of discotic liquid crystal phases--the nematics, chiral nematics, smectics, and the twist grain boundary phase (TGB). On Earth, the nucleation & crystal growth is affected by gravity due to differential density of the liquid crystal phase. So the space experiments are planned to observe the nucleation and crystal growth in microgravity. The effect of external force fields such as electric filed, magnetic field, temperature, and shear field on phase transitions is also planned to be studied in this project. To start with, we continued our studies by observing phase behavior of Zirconium Phosphate nanoplates exfoliated using Tertiary Butyl Ammonium Hydroxide suspensions. We tried to experimentally realize the I-N-N*-TGB-S phase transition which as per the simulations should to take place for the case of discotic liquid crystals with aspect ratio (thickness to diameter ratio) in the range 0.001 to 0.1. The observation of liquid crystal samples at various concentration between pair of cross polarizer gave a hint regarding existence of I, N, N*, TGB, and S phases. Techniques such as Circular Dichroism, Polarized Optical Microscopy, X-ray synchrotron were used to identify N*, TGB, and S phases and to establish the phase diagram. However as we plan to gather enough proofs for the existence of N*, TGB, & S phases in near future. In order to study non-charged platelets, we grafted poly(N-isopropylacrylamide) (PNIPAM) onto the zirconium phosphate (ZrP) platelets using pre-irradiated polymerization. The I-N transition was investigated by adjusting temperature for a single set of samples. We found that soft disks self-assemble into nematic liquid crystals in a wider thickness-over-diameter ratio than do hard disks. 2. Temperature-dependent I-N transition of charged nanoplates We plan to investigate liquid crystal behaviors in microgravity under a temperature gradient. So we study the temperature change effect on phase behavior in 1g first. We established the isotropic-to-nematic (I-N) transition phase diagram of charged platelets in the temperature-versus-volume fraction plane. We discover that the N phase can be melted by increasing temperature, and coexistent samples are more sensitive to polydispersity at higher temperature and higher concentration. 3. Effect of salt concentration on I-N coexistence of charged nanoplates We planned to use salt (TBA+Cl-) to tune the interaction between nanoplates. The effect of salt concentration is studied for the I-N phase transition and co-existence. Liquid crystals of different volume fraction of ZrP exfoliated using TBAOH at different salt concentration were prepared and their time dependent phase behavior was observed under the pair of cross polarizers. We chose the sample aged for sufficient time (10 days) in order to observe I-N coexistence. We find that the I-N coexistence region wider for salt concentration is higher. The effect of salt concentration on the chirality is yet to be fully understood in near future. We propose that the chirality should be reduced due to screening of colloidal interaction caused by the salt ions. Hence chiral phase at lower salt concentration will be developed at lower nanoplates volume fractions than those at higher salt concentration.
Bibliography Type:	Description: (Last Updated: 09/04/2025)
Abstracts for Journals and Proceedings	Cheng Z. "Colloidal Discotic Liquid Crystals." 29th American Society for Gravitational and Space Research and 5th International Symposium for Physical Sciences in Space, Orlando, FL, November 3-8, 2013. 29th American Society for Gravitational and Space Research and 5th International Symposium for Physical Sciences in Space, Orlando, FL, November 3-8, 2013. Program and abstracts, p. 117. https://asgsr.org/images/stories/pdf/Complete_Program_FINAL_2013_Meeting.pdf , Nov-2013
Abstracts for Journals and Proceedings	Cheng Z. "The Synthesis and Self-assembly of Layered Crystals." Presented at the AIChE Annual Meeting, Hilton San Francisco Union Square, San Francisco, CA. November 3-8, 2013. AICHE Annual Meeting 2013, San Francisco, CA. November 3-8, 2013. , Nov-2013
Abstracts for Journals and Proceedings	Cheng Z. "Nanoparticles as Models of Atoms and Building Blockers of Soft Matter." Presented at the Invited Seminar, Department of Chemical Engineering at the University of Florida, Nov. 18, 2013. Seminar, Department of Chemical Engineering at the University of Florida, Nov. 18, 2013. , Nov-2013
Abstracts for Journals and Proceedings	Cheng Z. "Nano Materials Research at TAMU." Presented at the Norway-Texas Nano Research Colloquium and Workshop on Nanoscience and Technology, NTNU Trondheim, Oct. 14-17, 2013. Norway-Texas Nano Research Colloquium and Workshop on Nanoscience and Technology, NTNU Trondheim, Oct. 14-17, 2013. , Oct-2013
Abstracts for Journals and Proceedings	Cheng Z, Wang X, Mejia A, Li H. "Thermal sensitive discotic liquid crystals." Presented at the 88th ACS Colloid and Surface Science Symposium, Philadelphia, Pennsylvania, June 22-25, 2014. 88th ACS Colloid and Surface Science Symposium, Philadelphia, Pennsylvania, June 22-25, 2014. , Jun-2014

Abstracts for Journals and Proceedings	Cheng Z. "Nanoplate Liquid Crystals." Presented at Liquid Crystals in Space workshop, 25th International Liquid Crystal Conference, Trinity College Dublin, Ireland, June 29-July 4, 2014. Liquid Crystals in Space workshop, 25th International Liquid Crystal Conference, Trinity College Dublin, Ireland, June 29-July 4, 2014. , Jun-2014
Articles in Peer-reviewed Journals	Wang X, Zhao D, Diaz A, Nava Medina IB, Wang H, Cheng Z. "Thermo-sensitive discotic colloidal liquid crystals." Soft Matter. 2014 Oct 21;10(39):7692-5. http://dx.doi.org/10.1039/c4sm00797b ; PubMed PMID: 25175847 , Oct-2014
Articles in Peer-reviewed Journals	Li HW, Wang X,Chen Y, Cheng Z. "Temperature-dependent isotropic-to-nematic transition of charged nanoplates." Physical Review E. 2014 Aug;90(2-1):020504. Published online 29 August 2014. http://dx.doi.org/10.1103/PhysRevE.90.020504 ; PMID: 25215677 (Note figure selected for display in kaleidoscope of the Journal.) , Aug-2014